

Identifying Predictors of User Adoption

Definition of Adoption

An **adopted user** is defined as a user who logged into the product on **three separate days within any seven-day period**. Login data was aggregated at the user-day level to avoid counting multiple logins on the same day.

Methodology

Two datasets were used: a user table containing signup attributes and an engagement table containing daily login activity. An adoption label was created for each user based on login behavior. Exploratory analysis was conducted to compare adoption rates across user attributes, followed by a **logistic regression model** using signup-time features to identify predictors of adoption. Model performance was evaluated using ROC-AUC.

Key Findings

Approximately **13% of users** met the adopted-user definition.

- **Signup method** was a strong predictor: users signing up via **Google Authentication** or **guest invitations** were more likely to adopt, suggesting lower friction and social onboarding increase engagement.
- **Invitation status** mattered: users invited by existing users showed higher adoption rates, reinforcing the importance of collaborative context.
- **Organization size** showed a nonlinear relationship: users in **medium-sized organizations** adopted more than those in very large organizations.
- **Personal project invitations** were strongly associated with lower adoption, indicating more exploratory or short-term usage.
- **Email domain** had some predictive power, likely acting as a proxy for account authenticity.
- Marketing email opt-in and drip campaigns showed **minimal impact** on adoption.

Model Results & Limitations

The logistic regression model achieved a **ROC-AUC of ~0.63**, indicating a meaningful predictive signal given the limited feature set. The analysis is limited to signup-time attributes and does not incorporate product usage, collaboration behavior, or session-level metrics. Including early usage and team interaction data would likely improve predictive performance.